# Robotics Academy: New Power Tower Inspection Using Deep Learning

Program: GSoC 2026

Organization: JdeRobot

Project: #3

Project Size: Medium (~175h)

Duration: Jun – Sep 2026


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Demo Video: https://youtu.be/aZTyB_sHlaI

Time Zone: UTC+3

Mentors**:** David Pascual-Hernández, Md. Shariar Kabir, Luis Roberto Morales

## 1. ABSTRACT


Power infrastructure inspection is a safety-critical task typically performed manually at significant cost and risk. This project proposes a new deep learning-based exercise in JdeRobot's RoboticsAcademy where students train and deploy a neural network that detects and classifies structural defects on power towers using a simulated drone as the inspection platform. Key outcomes include a Gazebo Harmonic simulation environment, a labeled defect dataset, a reference YOLOv8 model, an extended Simple API, and comprehensive Jupyter Notebook learning materials.


## 2. MOTIVATION

RoboticsAcademy already hosts a classical power-tower inspection exercise relying on colour thresholds and edge detection. While useful, it does not teach students to apply modern deep learning pipelines. Three problems motivate this project: - No deep learning exercise exists connecting drone perception with structured defect classification. - Students lack a reproducible simulation environment and labelled dataset to train their own models. - The existing Simple API has no standardised way to return structured DL outputs (bounding boxes, logits, confidence).

# 3. TECHNICAL PROPOSAL

## 3.1 Simulation Environment

A Gazebo Harmonic scene featuring power towers with physically modelled defects (corrosion, broken insulators, missing hardware). A drone with RGB camera will navigate pre-programmed waypoints — mirroring real UAV inspection workflows.

## 3.2 Dataset Collection & Labelling

Images recorded from Gazebo across multiple configurations and lighting conditions. Bounding-box annotations produced using CVAT, exported in YOLO and COCO formats. Dataset uploaded to HuggingFace Hub or Zenodo.

## 3.3 Deep Learning Model

Reference model trained using YOLOv8 (primary) and EfficientDet (secondary). Training tracked with MLflow. Target: mAP@0.5 > 0.75. Trained weights distributed alongside the exercise.

## 3.4 Exercise Integration

Follows architecture of existing End-to-End Visual Control exercise. Students implement a Python module with execute() function receiving a camera frame and returning structured predictions. RADI Docker image extended with new Gazebo world and HAL/GUI bindings.

## 3.5 Simple API Extension

Four new output fields to strictly follow JdeRobot's requirements:

- set_boxes(boxes) — bounding box coordinates
- set_labels(labels) — integer class indices
- set_logits(logits) — raw unnormalized predictions for classification
- set_scores(scores) — confidence values [0,1]

## 4. TIMELINE

Week 1-2 | Community Bonding — study codebase, mentor alignment

Week 3-4 | Gazebo power tower scene + drone navigation

Week 5-6 | Record dataset + label with CVAT

Week 7-8 | Train YOLOv8 model, evaluate with mAP/F1

Week 9 | MIDTERM — working defect detector + metrics

Week 10-11 | RoboticsAcademy exercise integration

Week 12-13 | Simple API extension (bbox, logits, confidence)

Week 14 | Jupyter Notebooks + dataset upload

Week 15-16 | Testing, final polish, submission

## 5. DELIVERABLES

1) Gazebo Harmonic power tower simulation environment
2) 500+ labeled image dataset (public HuggingFace/Zenodo)
3) Trained YOLOv8 model (mAP > 0.75)
4) Extended Simple API with bbox, logits + confidence support
5) Fully integrated RoboticsAcademy exercise
6) Jupyter Notebook tutorials and documentation

## 6. TOOLS & TECHNOLOGIES

Simulation: Gazebo Harmonic (or Classic, pending mentor alignment), ROS2 Humble, Aerostack2

Deep Learning: Python, PyTorch, YOLOv8, EfficientDet

Data: CVAT, Roboflow, HuggingFace

Tracking: MLflow

Infrastructure: Docker, HAL/GUI API, GitHub Actions

## 7. ABOUT ME

I am Mustafa Kerem Güçlü, a 2nd-year Computer Engineering student at Ankara University. During this GSoC application I successfully set up the RoboticsAcademy Docker environment and implemented a Follow Line solution featuring adaptive PID control with dynamic Kp adjustment, asymmetric brake/acceleration system, and real-time HUD visualization. Why this project? Power tower inspection sits at the intersection of drone robotics, computer vision, and deep learning — the exact areas I want to deepen. I am motivated by the social impact of safer infrastructure inspection and by creating learning materials for future engineering students. Commitment: I can dedicate 25-30 hours per week throughout GSoC. No conflicting internships or exams during the coding phase.

Demo Video: https://youtu.be/aZTyB_sHlaI

GitHub: [github linkin]

## 8. RISKS & MITIGATION

Gazebo complexity → Use simplified procedural towers, start data collection early

Gazebo version mismatch → Will align with mentors on whether to use Gazebo Harmonic or stick to the current Gazebo Classic baseline during community bonding.

Model below target → Fall back to pretrained COCO backbone, transfer learning

Docker integration issues → Follow existing exercise PRs, consult mentors early